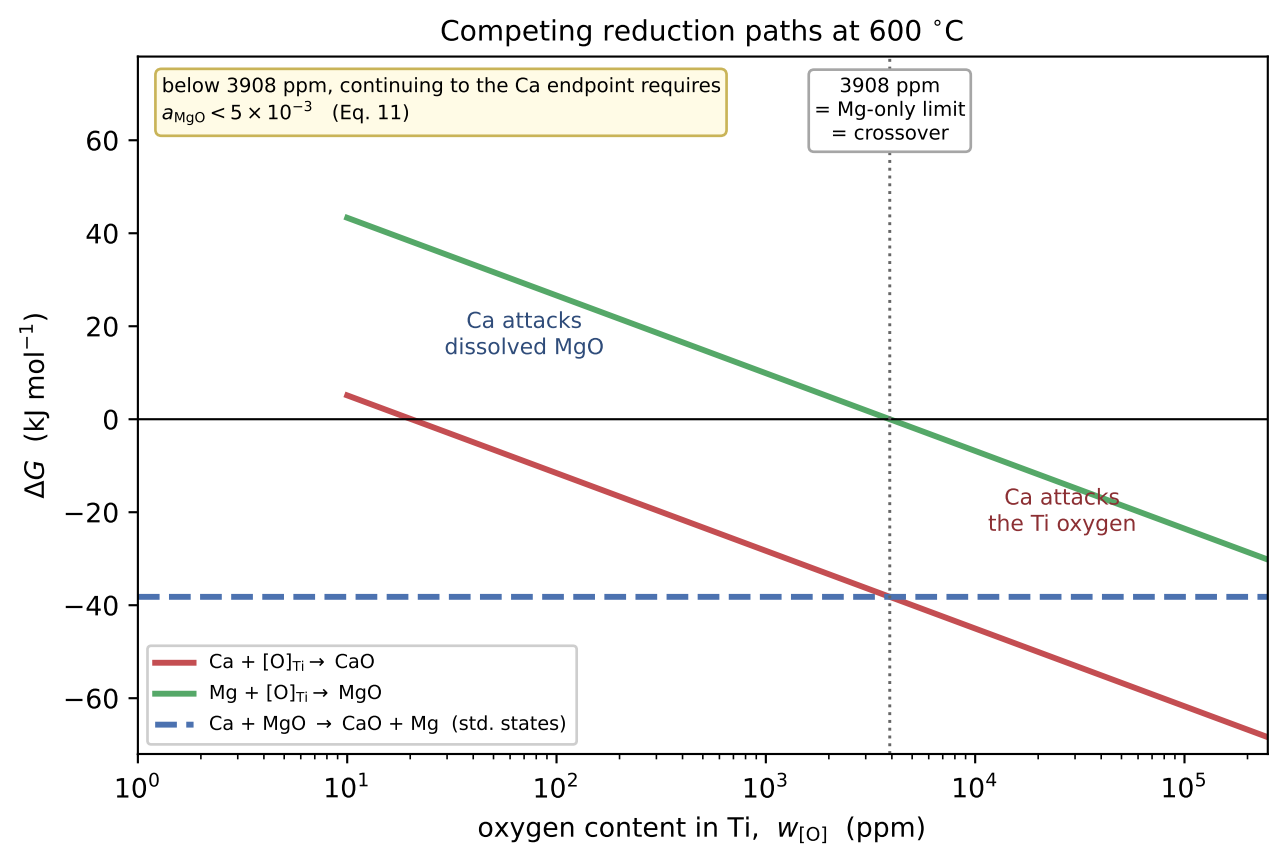
Competing reduction paths at 600 °C
below 3908 ppm, continuing to the Ca endpoint requires
$a_{MgO} < 5 \times 10^{-3}$ (Eq. 11)
3908 ppm
= Mg-only limit
= crossover
Ca attacks
dissolved MgO
Ca attacks
the Ti oxygen
Ca + [O]$_{Ti}$ → CaO
Mg + [O]$_{Ti}$ → MgO
Ca + MgO → CaO + Mg (std. states)
ΔG (kJ mol^{-1})
oxygen content in Ti, $w_{[O]}$ (ppm)
60
40
20
0
−20
−40
−60
10^0
10^1
10^2
10^3
10^4
10^5